Mehul Goel

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Education

Carnegie Mellon University - School of Computer Science Pittsburgh, PA
BS in Computer Science, Minor in Machine Learning | ***High Honors Dean's List Every Semester*** *May 2026*
Courses**: DL Systems | Deep Reinforcement Learning | Intro to Deep Learning | Data Structures and Algorithms* ***4.0 GPA

Experience

Software Developer Intern May 2025 – August 2025
Hudson River Trading *New York City, NY*
- Created full stack language server from scratch, using a microservice based protocol implemented in pygls
- Optimized parser by 50% for large files by refactoring common functions into C++ using nanobind
- Modularized numerous market simulators into one product, re-implementing markets like NASDAQ & NYSE Pillar
- Refactored market protocol logic by utilizing C++ reflections to generalize packet structures across exchanges

Robotics Research Intern August 2024 – May 2025
Carnegie Mellon University Robotics Institute *Pittsburgh, PA*
- Rigid body manipulation with quadruped using Drake's solver for future work in imitation learning
- Studied 3-axis legged kinematics and dynamics for flipping and spinning boxes using quadruped robot
- Reduced error by 20% in real-world state estimator based on multi-camera state estimation and Kalman Filtering

MattLab - Computer Vision ML Research Intern August 2023 – May 2024
Carnegie Mellon University Robotics Institute ○ Apple Inc. *Pittsburgh, PA*
- Developed real-time image segmenter for internals of iPhones and iPads based on ResNet50 model
- Classified electronic parts achieving an IOU (Accuracy Metric) of 96% after upgrading to a SORT tracking solution
- Detected screws with 93% accuracy on Apple devices after incorporating a top-down YOLO V8 model

Introduction to Imperative Programming Teaching Assistant January 2024 – May 2025
Carnegie Mellon University *Pittsburgh, PA*
- Leading interactive labs for 60+ students, teaching fundamentals of coding, C, and data structures
- Working with professors to create collaborative and engaging problem sets to challenge all 500 students
- Held office hours for personalized support, focusing on conceptual questions and complex C programs

Projects

RoboBuggy Software Lead | *Autonomous Driving, ROS, Python, LiDAR Vision* August 2023 – Present
- Developed, tested, and raced the perception, controls, vision software stack for 30mph autonomous vehicle
- Created a perception system with 10m range and 5% variance using light-invariant stereo camera and OpenCV
- Implemented MPC based non-linear controller for autonomous driving using ROS to control hardware
- Explored reinforcement learning based solutions to obstacle avoidance, including end-to-end models

GPT2 Recreation | *Deep Learning, PyTorch, NumPy, LLMs, Transformers* June - August 2024
- Created a character prediction model, based on GPT2 Architecture, following Andrej Karpathy's tutorials
- Learned DL basics for GPT2 by following along on Stanfords 231N Summer course on Transformers, RNNs, LSTMs
- Explored speech transformer with character error rate of 22.8, using BiLSTM Embedding (Dong et al. 2018)

MBR Sim Author | *Python, ResNet, BERT, GPT, Transformers, Matplotlib* September 2022 – Feb 2023
- Co-authored paper discussing software-based silicon performance modeling for ML models like BERT & GPT
- Within 3% of real-life results for power and heat simulations over 10+ ML workloads on CPUs, GPUs, and TPUs
- Developed memory modelling software, including hardware resource manager, for various silicon products

GlobaLex | *Microsoft Azure, ReactJS, Python, GPT4, Whisper, MongoDB* February 2024
- Developed real-time translation web-calling software for MS Imagine Cup and won $5,000 in tech credits
- Utilized OpenAI's Whisper API to translate between English and Spanish in real-time on 30 second clips
- Built front-end with ReactJS and MongoDB including authentication and performed web calls using Azure

Technical Skills

Languages: Python, C++/C, OCAML/SML, Java, JavaScript/TypeScript, Julia, Assembly x86/64, SQL, Rust, Go, Dart
Frameworks: PyTorch, ROS, ReactJS, NodeJS, OpenCV, Tensorflow, Tailwind, Azure, Git, Firebase, Git, Docker, LaTeX